ELUMALAI S

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Summary

Full Stack Developer with over 3.8 years of experience in software development, research and development (R&D), machine learning, and cloud-based solutions. Experienced in designing and delivering scalable web and mobile applications using Django, Angular, React, and Flutter. Skilled in AWS cloud services, Docker-based deployments, Agile development practices, backend architecture, API integrations, and quality assurance. Proven track record of leading R&D initiatives, developing AI-driven solutions, and translating innovative concepts into production-ready applications.

Work Experience

- **Tender Software India Private Limited**
  Software Engineer: **Jan 2026 – Present**
  - Developed backend services using Python and FastAPI, building scalable REST APIs for healthcare and enterprise applications.
  - Implemented 3D mesh processing pipelines using Trimesh and Open3D to analyze PLY scan data and extract limb measurements such as height and circumference.
  - Designed and optimized PostgreSQL database schemas and APIs for managing users, patients, scans, and clinical records.
  - Built multi-tenant backend architecture supporting entity-level access control with Super Admin and role-based permissions.
  - Implemented image upload and medical data pipelines, including conversion of scan images to DICOM format for medical imaging compatibility.
  - Deployed and managed backend services on AWS EC2 and S3, with version control and collaboration using Git.
- **Techvivant**
  Python Full Stack Developer: **Jul 2025 – Dec 2025**
  - Contributing to the development and optimization of the **Jtheta.ai** medical annotation platform.
  - Implemented auto-save functionality for real-time annotation updates, improving usability and reliability.
  - Integrated **FastAPI** microservices with **Django** to enhance performance and modularity.
  - Optimized backend and frontend workflows to handle large medical datasets efficiently.
  - Enhanced 3D and 2D visualization using **OpenCV** and **Three.js** for medical image projections.
  - Implemented **Redis** caching and optimized database queries to improve response time and scalability.
  - Provided production support, debugging issues, and deploying updates in live environments.
  - Collaborated with the R&D team to enhance annotation workflows and model training pipelines.
- **Brain Insight**
  Python Full Stack Developer: **Jan 2023 – July 2025**
  - Contributed to the development and enhancement of software solutions.
  - Wrote clean, maintainable code; debugged issues; and implemented 10+ new features.
  - Collaborated with cross-functional teams to meet project milestones.
  - Participated in Agile ceremonies including daily stand-up's, sprint reviews, and retrospectives.

  - Assisted in integrating REST APIs and supported CI/CD deployment pipelines.

- **Atos Syntel**
  Trainee Web Developer (Associate Consultant): **Sept 2022 – Dec 2022**
  - Assisted in developing and maintaining web applications.
  - Fixed bugs, wrote modular code, and supported senior developers.
  - Gained real-time project experience in frontend/backend development.
  - Learned web technologies and Agile workflows through guided training.

Projects

- **Stocking Scanner**
  Tech Stack: Python, FastAPI, AWS EC2, AWS S3, PostgreSQL, Git.
  Role: Software Engineer
  - Developed backend APIs for user management with Super Admin and User roles.
  - Implemented PLY file processing using Trimesh and Open3D for handling 3D scan meshes.
  - Built logic to analyze 3D mesh data for limb measurements such as height and circumference.
  - Designed APIs for scan data upload, processing, and patient record management.
  - Implemented audit logging and soft delete functionality to maintain data traceability.
  - Implemented secure report delivery via Gmail SMTP, enabling automated email distribution of patient reports through password-protected and secure download workflows.
- **Edema Tracker**
  Tech Stack: Python, FastAPI, AWS EC2, AWS S3, MySQL, Git.
  Role: Software Engineer
  - Developed backend APIs for patient edema monitoring and scan management.
  - Implemented image upload functionality for storing patient scan images.
  - Built pipeline to convert uploaded images into DICOM format for medical imaging compatibility.
  - Designed database structures for patient records, scan history, and measurement tracking.
  - Optimized data retrieval and storage for medical scan records.
- **Blue dop**
  Tech Stack: Python, FastAPI, AWS EC2, AWS S3, MySQL, Git.
  Role: Software Engineer
  - Developed backend system using multi-tenant architecture to support multiple entities in a single platform.
  - Implemented Entity and Super Admin role management with permission-based access control.
  - Built APIs for entity, clinic, user, and patient management.
  - Optimized patient lookup and data retrieval performance for large datasets.
  - Implemented audit logging and activity tracking for system operations.
  - Designed and integrated AWS Cognito-based Multi-Factor Authentication (MFA) supporting TOTP, SMS, and Email authentication methods with configurable primary and secondary MFA workflows.
  - Developed custom AWS Lambda functions to integrate WhatsApp OTP authentication using Meta WhatsApp Business APIs, enabling secure multi-channel user verification and authentication.

- **Jtheta.ai**
  Role: Python Full Stack Developer
  Tech Stack: Python, Django, FastAPI, PostgreSQL, AWS EC2, S3, Redis, JavaScript, OpenCV, Three.js
  Methodology: Agile
  - Contributed to the development of **Jtheta.ai**, a multi-domain annotation platform supporting medical, LiDAR, satellite, and general image annotations.
  - Enhanced backend performance by integrating **FastAPI** with **Django**, enabling faster data processing and real-time annotation updates.
  - Worked extensively on medical annotation modules, adding slicing, 3D visualization, and auto-save features to improve usability.
  - Implemented **Redis** caching and optimized memory management for large dataset handling.
  - Integrated **HubSpot** tracking for user activity insights and configured **AWS S3** with **PostgreSQL WAL** backups for secure data storage and recovery.
  - Provided production support, fixing critical bugs and improving overall system stability across annotation workflows.
- **USA Medical Review Portal**
  Role: Python Full Stack Developer
  Tech Stack: Python (AWS SDK), React.js, .NET, AWS CDK
  Methodology: Agile
  - Contributed as a core developer to a USA-based medical review portal fully developed using AWS CDK for infrastructure as code.
  - Developed and maintained backend services using Python integrated with AWS SDK, and implemented dynamic, responsive UI components with React.js.
  - Integrated a .NET-based bulk review upload feature as part of the system's high-volume data processing module.
  - Followed Agile methodology, participating in sprint planning, reviews, retrospectives, and daily standups to ensure continuous delivery and iterative improvement.
  - Designed and executed comprehensive Unit, Integration, Smoke, System, and Feature Branch Testing, ensuring robustness and reliability throughout the deployment lifecycle.
  - Played a key role in architecture discussions, contributing to both cloud infrastructure and application-level decisions.
- **Next Generation Soil Moisture Prediction**
  Role: Junior Full Stack Developer
  Tech Stack: Django, Angular, LSTM, AWS
  Methodology: Agile
  - Developed a full-stack web application using Django for the backend and Angular for the frontend as part of an organizational research project.
  - Designed and implemented a soil moisture prediction model using the LSTM (Long Short-Term Memory) algorithm based on custom datasets.
  - Actively managed weekly client requirements and feature updates, ensuring smooth delivery through iterative Agile practices.
  - Deployed the project on AWS, including provisioning and managing a dedicated EC2 server for organizational use.
- **Object Detection Projects**
  Role: Junior Computer Vision Engineer
  Tech Stack: Roboflow, YOLOv5, YOLOv8, Python
  Methodology: Experimental/Agile

- Collected and annotated datasets using Roboflow for two key use cases:
    - Pig Detection: Trained the dataset using YOLOv5 to detect moving pigs.
    - Bridge Crack Detection: Trained a crack dataset using YOLOv8 for structural safety analysis.
- Performed model evaluation and iterative training to improve accuracy and performance.
- Followed a rapid prototyping and testing cycle aligned with Agile experimentation.

Skills

- **Languages:** Python, Dart, SQL
- **Markup Languages:** HTML, CSS
- **Cloud Technologies:** AWS (EC2, S3, ECS, Lambda, IAM, Code Pipeline, Cognito)
- **Development Tools:** VS Code, Android Studio, Jupyter Notebook
- **Project Management Tools:** JIRA, Trello
- **Machine Learning:** Random Forest, Decision Tree, SVM, XGBoost
- **Frameworks:** Angular, Flask, Django, React, Flutter, FastAPI
- **Operating Systems:** Windows, Ubuntu, Amazon linux
- **Version Control & Tools:** Git & GitHub, Docker, Redis
- **Top 5 Key Skills:** Python, Django, FastAPI, AWS, Redis

Education

- **BE-CSE [Bachelors of Computer Science]** 2018–2022
  Kingston Engineering College - Vellore - 7.27 CGPA

Declaration

I hereby declare that the particulars stated above are true to the best of my knowledge and belief.